

Bekretsion Seyoum

Backend Software Engineer

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SUMMARY

Backend software engineer building voice AI platforms, APIs and automation pipelines with Node.js, TypeScript and SQL. Software Engineer at Pyronix AI since 03/2025; national finalist, ALX Ethiopia x Kuriftu Hospitality Hackathon 2026.

EXPERIENCE

Software Engineer - Pyronix AI

Remote | 03/2025 - Present

- Full-stack development with Node.js at a custom AI and software development company that builds for US clients.

Intern - Space Science and Geospatial Institute (SSGI)

Addis Ababa, Ethiopia | 06/2026 - 09/2026

- Built a hybrid CNN-LSTM model in Python that forecasts the Dst geomagnetic storm index from NASA OMNI2 solar-wind data.
- Rebuilt the pipeline to predict the change in Dst from physics-based features; the second version beat the persistence baseline at every forecast horizon (6 hours ahead: 7.03 nT mean absolute error versus 8.08 nT).
- Traced the first version's apparent failure to learn to an early-stopping setting, not the model, and corrected the write-ups.

PROJECTS

Hello AI - AI voice receptionist platform

National Finalist, ALX Ethiopia x Kuriftu Hospitality Hackathon 2026 | hello.bekretsion.com

Node.js, Express, MySQL, ElevenLabs, Vapi, Stripe, Dropbox Sign, Google Calendar, OAuth 2.0

- Designed a dual-provider voice architecture (ElevenLabs and Vapi) behind one interface; switching providers is one environment variable.
- Implemented Stripe webhooks with raw-body HMAC signature verification, per-minute billing and automatic minute top-ups.
- Built a post-call automation engine (extract, map, dispatch) that sends results to Slack, Outlook and CRMs without touching the core call pipeline.
- 85+ sequential database migrations run on startup; supports 95+ languages, including Amharic, Afaan Oromo and Tigrinya.

Lead Qualification Automation

n8n, Node.js, PostgreSQL, Google Gemini, HubSpot, Slack | bekretsion.com/projects/lead-qualification

- Inbound-lead pipeline: the webhook acknowledges in about 70 ms, then validates, de-duplicates, classifies each lead with an LLM and scores it in code.
- In live testing the sales rep was notified in 6.6 s and the prospect received a personalised reply in 8.1 s; 11 of 11 test leads were routed correctly.
- 25 simultaneous identical submissions produced one contact (idempotency key with a UNIQUE constraint); a dead-letter queue with scheduled replay meant 0 leads lost.

Document Invoice Processing

n8n, Node.js, Tesseract.js OCR, LLM field extraction | bekretsion.com/projects/invoice-processing

- Extracts invoice fields from native, scanned and photographed invoices with OCR and an LLM, and re-checks every amount in integer minor units.
- Posts an invoice only when its numbers reconcile; tested against 16 fixtures built to break it, including broken arithmetic, unknown vendors and changed bank details.

EDUCATION

BSc Computer Science - Hope Enterprise University College

Addis Ababa, Ethiopia | Expected graduation 07/2027

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend and web: Node.js, Express, Prisma, REST APIs, WebSockets, webhooks, Next.js, React

Databases: PostgreSQL, MySQL, Redis

Auth and security: JWT (RS256), OAuth 2.0, RBAC, HMAC webhook verification, bcrypt

AI, automation and integrations: n8n, Google Gemini, ElevenLabs, Vapi, Tesseract.js OCR, Stripe, Dropbox Sign, Google Calendar, Microsoft Graph, HubSpot, Slack

Infrastructure: Docker, GitHub Actions, Vercel, Render, Neon, Upstash